
Q-Drop Quantum GCN — Full Training Pipeline & Learning Rate Schedule
Q-Drop Training Pipeline — IMDB-Binary
FORWARD PASS
Data
IMDB-Binary
1 000 ego-graphs
10-Fold CV
Degree One-Hot
136-dim node features
(clamped deg ≤ 135)
DataLoader
batch_size=8
shuffle=True (train)
QGCN
QGCNConv × 2
n_qubits=16
Global Mean Pool
graph-level embedding
dim=16
Linear Classifier
16 → 1
BCEWithLogitsLoss
Loss
class-weighted
pos_weight
Optim
Early Stop
patience=15
best-state restore
Q-Drop
accumulate/prune
wire masking
OneCycleLR
cos anneal
pct_start=0.1
AdamW
2 param groups
weight_decay=1e-3
Grad Clip
max_norm=1.0
clip_grad_norm_
Backward
autograd
BACKWARD PASS
OneCycleLR Schedule — Classical vs Quantum Param Groups (100 Epochs)
warmup
end
(10%)
Classical params
Quantum params (×0.1 scale)
peak=2.50e-03
init=2.50e-03
min=5.0e-05
Learning Rate
Epoch
LR Calculation — Key Values
OneCycleLR Parameters
base_lr = 5e-03
pct_start = 0.1
div_factor = 10
final_div_factor = 50
anneal_strategy = 'cos'
Classical group
max_lr = base_lr × 5 = 0.0250
init_lr = max_lr ÷ 10 = 0.0025
min_lr = init_lr ÷ 50 = 5.0e-05
Quantum group
scale = quantum_lr_scale = 0.1
max_lr = 0.0250 × 0.1 = 2.5000e-03
init_lr = 2.50e-03 ÷ 10 = 2.50e-04
min_lr = 2.50e-04 ÷ 50 = 5.00e-06
total_steps
= epochs × steps_per_epoch
= 100 × ~113 = 11,300
warmup = 1,130 steps (10%)
anneal = 10,170 steps (90%)
Warmup Phase (0 → 10%)
linear
warm-up
Classical
Quantum
warmup zone
Learning Rate
Step
Cosine Annealing Formula
$lr(t) = lr_{min} + \frac{1}{2}(lr_{max} - lr_{min})(1 + \cos(\pi t))$
0.025
5.0e-05
Classical
Quantum
Learning Rate
Progress in anneal phase (t / T_anneal)
AdamW Update Rule
Step t (per param group):
g_t = ∇L (+ grad_clip ≤ 1.0)
m_t = β₁·m_{t-1} + (1-β₁)·g_t
v_t = β₂·v_{t-1} + (1-β₂)·g_t²
m̂_t = m_t / (1 - β₁ᵗ) [bias corr]
v̂_t = v_t / (1 - β₂ᵗ) [bias corr]
θ_t = θ_{t-1}
− lr(t) · m̂_t / (√v̂_t + ε)
− lr(t) · λ · θ_{t-1}
↑ weight_decay=1e-3
Defaults: β₁=0.9 β₂=0.999 ε=1e-8
Two lr values from OneCycleLR:
Classical: 0.0025 → 0.0250 → 5.0e-05
Quantum: 2.50e-04 → 2.50e-03 → 5.0e-06
Q-Drop Phase Cycle (accumulate_window=10 | prune_window=8 | prune_ratio=0.8)
▼ = wire-drop event
ACCUM
PRUNE
Classical LR (norm)
Quantum LR (norm)
Training Step (batches)
10-Fold CV + Early Stopping
F1 ep97
F2 ep78
F3 ep81
F4 ep94
F5 ep75
F6 ep87
F7 ep90
F8 ep53
F9 ep43
F10 ep58
Training
Skipped (early stop)
Best val stop
val_frequency=5 | patience=15 | best-state restored